Q16 At equilibrium,

$$F_E = \frac{qV}{d} = mg$$

When d is twice, F_E is reduced by half,

$$F_E' = \frac{1}{2}mg$$

$$F_{resultant} = mg - \frac{1}{2}mg = 0.5mg = ma$$

$$a = 0.5g$$

Oil drop accelerates downwards as weight is now greater than upward electric force.

Answer: C

Q17 The forces due to the charges at different positions are shown